.1 A Hull, Deck, Navigation and Safety services;
.2 B Propulsion service auxiliaries;
.3 C Auxiliary Engine and Main Engine Services;
.4 D Ship's General services;

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where load reductions might be possible. The groups are listed below:

.1 A Hull, Deck, Navigation and Safety services;
.2 B Propulsion service auxiliaries;
.3 C Auxiliary Engine and Main Engine Services;
.4 D Ship's General services;
.5 E Ventilation for Engine-rooms and Auxiliaries room;
.6 F Air Conditioning services;
.7 G Galleys, refrigeration and Laundries services;
.8 H Accommodation services;
.9 I Lighting and socket services;
.10 L Entertainment services;
.11 N Cargo loads; and
.12 M Miscellaneous.

All the ship's loads have to be delineated in the document, excluding only PAeff, the shaft motors and shaft motors chain (while the propulsion services auxiliaries are partially included below in paragraph 4.1.2 B). Some loads (i.e. thrusters, cargo pumps, cargo gear, ballast pumps, maintaining cargo, reefers and cargo hold fans) still are included in the group for sake of transparency, however their service factor is zero in order to comply with rows 4 and 5 of paragraph 2.5.6 of the Guidelines, therefore making it easier to verify that all the loads have been considered in the document and there are no loads left out of the

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and 5 of paragraph 2.5.6 of the Guidelines, therefore making it easier to verify that all the loads have been considered in the document and there are no loads left out of the measurement.

4.1.1 A Hull, Deck, Navigation and safety services

.1 loads included in the Hull services typically are: ICCP systems, mooring equipment, various doors, ballasting systems, Bilge systems, Stabilizing equipment, etc. Ballasting systems are indicated with service factor equal to zero to comply with row 5 of paragraph 2.5.6 of the Guidelines;

.2 loads included in the deck services typically are: deck and balcony washing systems, rescue systems, cranes, etc.;

.3 loads included in the navigation servic es typically are: navigation systems, navigation's external and internal communication systems, steering systems, etc.; and

.4 loads included in the safety services typically are: active and passive fire systems, emergency shutdown systems, public address systems, etc.

4.1.2 B Propulsion service auxiliaries
This group typically includes: propulsion secon dary cooling systems such as LT cooling

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systems, emergency shutdown systems, public address systems, etc.

4.1.2 B Propulsion service auxiliaries
This group typically includes: propulsion secon dary cooling systems such as LT cooling pumps dedicated to shaft motors, LT cooling pumps dedicated to propulsion converters, propulsion UPSs, etc. Propulsion service Loads do not include shaft motors (PTI(i)) and the auxiliaries which are part of them (shaft motor own cooling fans and pump, etc.) and the shaft motor chain losses and auxiliaries which are part of them (i.e. shaft motor converters including relevant auxiliaries such as converter own cooling fans and pumps, shaft motor

transformers including relevant auxiliaries losses such as propulsion transformer own cooling fans and pumps, shaft motor Harmonic filter including relevant auxiliaries losses, shaft motor excitation system including the relevant auxiliaries consumed power, etc.). Propulsion

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fans and pumps, shaft motor Harmonic filter including relevant auxiliaries losses, shaft motor excitation system including the relevant auxiliaries consumed power, etc.). Propulsion service auxiliaries include manoeuvring propul sion equipments such as manoeuvring thrusters and their auxiliaries whose service factor is to be set to zero.

4.1.3 C Auxiliary Engine and Main Engine Services
This group includes: cooling systems, i.e. pumps and fans for cooling circuits dedicated to alternators or propulsion shaft engines (seawater, technical water dedicated pumps, etc.), lubricating and fuel systems feeding, transfer, treatment and storage, ventilation system for combustion air supply, etc.

4.1.4 D Ship's General services
This group includes Loads which provide general services which can be shared between shaft motor, auxiliary engines and main engine and accommodation support systems. Loads typically included in this group are: Cooling systems, i.e. pumping seawater, technical water main circuits, compressed air systems, fresh water generators, automation systems, etc.

4.1.5 E Ventilation for Engine-rooms and Auxiliaries room

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typically included in this group are: Cooling systems, i.e. pumping seawater, technical water

main circuits, compressed air systems, fresh water generators, automation systems, etc.

4.1.5 E Ventilation for Engine-rooms and Auxiliaries room
This group includes all fans providing ventila tion for engine-rooms and auxiliary rooms that typically are: Engine-rooms cooling supply-ex haust fans, auxiliary rooms supply and exhaust fans. All the fans serving accommodation areas or supplying combustion air are not included in this group. This group does not include cargo hold fans, and garage supply and exhaust fans.

4.1.6 F Air Conditioning services
All Loads that make up the air conditioning service that typically are: air conditioning chillers, air conditioning cooling and heating fluids transfer and treatment, air conditioning's air handling units ventilation, air conditioning re-heating systems with associated pumping, etc. The air conditioning chillers service factor of load, service factor of time and service factor of duty are to be set as 1 (kl=1, kt=1 and kd=1) in order to avoid the detailed validation of the

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The air conditioning chillers service factor of load, service factor of time and service factor of duty are to be set as 1 (kl=1, kt=1 and kd=1) in order to avoid the detailed validation of the heat load dissipation document (i.e. the chiller's electric motor rated power is to be used). However, kd is to represent the use of spare chillers (e.g. four chillers are installed and one out four is spare then kd=0 for the spare chiller and kd=1 for the remaining three chillers), but only when the number of spare chillers is cl early demonstrated via the heat load dissipation document.

4.1.7 G Galleys, refrigeration and Laundries services
All Loads related to the galleys, pantries refrigeration and laundry services that typically are: Galleys various machines, cooking appliances, galleys' cleaning machines, galleys auxiliaries, refrigerated room systems including refrigeration compressors with auxiliaries, air coolers, etc.

4.1.8 H Accommodation services
All Loads related to the accommodation services of passengers and crew that typically are: crew and passengers' transportation systems, i. e. lifts, escalators, etc., environmental

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coolers, etc.

4.1.8 H Accommodation services
All Loads related to the accommodation services of passengers and crew that typically are: crew and passengers' transportation systems, i. e. lifts, escalators, etc., environmental services, i.e. black and grey water collecting, transfer, treatment, storage, discharge, waste systems including collecting, transfer, treat ment, storage, etc., accommodation fluids transfers, i.e. sanitary hot and cold water pu mping, etc., treatment units, pools systems, saunas, gym equipments, etc.

RESOLUTION MEPC.212(63)
Adopted on 2 March 2012
2012 GUIDELINES ON THE METHOD OF CALCULATION OF THE
ATTAINED ENERGY EFFICIENCY DESIGN INDEX (EEDI) FOR NEW SHIPSMEPC 63/23

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4.1.9 l Lighting and socket services
All Loads related to the lighting, entertainment and socket services. As the quantity of lighting circuits and sockets within the ship may be significantly high, it is not practically feasible to list all the lighting circuits and points in the EPT for EEDI. Therefore circuits should be grouped into subgroups aimed to identify possible improvements of efficient use of

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feasible to list all the lighting circuits and points in the EPT for EEDI. Therefore circuits should be grouped into subgroups aimed to identify possible improvements of efficient use of power. The subgroups are:

.1 Lighting for 1) cabins, 2) corridors, 3) technical rooms/stairs, 4) public spaces/stairs, 5) engine-rooms and auxiliaries' room, 6) external areas, 7) garages and 8) cargo spaces. All have to be divided by main vertical zone; and

.2 Power sockets for 1) cabins, 2) corridors, 3) technical rooms/stairs,